paper_id: "PAPER_1107"
title: "UQFF 26D Geometric Folding Operator and Folded Spacetime Metric"
session: 225
date: 2026-04-16
author: "Daniel T. Murphy"
status: production
cvw: "v2.0.0"
tags: [26D, geometric-folding, conformal-factor, Wolfram-hypergraph, Schwarzschild, folded-metric, quantum-layers, U
crosslinks: [PAPER_624, PAPER_1100, PAPER_1106]
sm_anchor: "CVW v2.0.0 – G6 SM Anchor Gate compliant"

PAPER_1107: UQFF 26D Geometric Folding Operator and Folded Spacetime Metric

Abstract

We derive the UQFF 26D geometric folding operator $\mathscr{F}_{26}(x)$ that maps fields from the full 26-dimensional quantum layer space to effective 4D spacetime, and construct the folded metric $ds^2_{\text{folded}} = g_{\mu\nu} \cdot \mathscr{F}_{26}(r)$. The folding parallels Wolfram hypergraph rewriting, with each quantum layer as a node and phonon resonances as edges.

1. The 26D Folding Operator

$$\mathscr{F}_{26}(x) = x \cdot (26!)^{-1/13} \cdot S_{26}^{(3)}([\text{SSq}]) \cdot \Phi_{1.25\,\text{THz}}(\omega, \Gamma)$$

Folding Normalization

The factorial scaling $(26!)^{-1/13}$ arises from the combinatorial weight of folding 22 compact dimensions into 4 extended ones:

$$26! = 4.032914611 \times 10^{26}$$

$$(26!)^{-1/13} \approx 1.176 \times 10^{-2}$$

This ensures the folded field has the correct dimensional reduction weight.

2. Folded Spacetime Metric

Starting from a Schwarzschild base in 4D:

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) dt^2 + \frac{dr^2}{1 - r_s/r} + r^2\, d\Omega^2$$

The folded metric applies $\mathscr{F}_{26}$ as a conformal factor:

$$ds^2_{\text{folded}} = \mathscr{F}_{26}(r) \cdot \left[-\left(1 - \frac{r_s}{r}\right) dt^2 + \frac{dr^2}{1 - r_s/r} + r^2\, d\Omega^2\right]$$

Effective Schwarzschild Radius

$$r_{s,\text{folded}} = r_s \cdot (26!)^{-1/13} \cdot S_{26}^{(3)} \cdot \Phi_{\text{gauss}}$$

The folding operator suppresses the effective gravitational radius by the combined phonon-buoyancy factor.

3. Layer-by-Layer Folding

Each of the 26 quantum layers contributes with its own state factor Q_i:

$$Q_i = \frac{1}{1 + \kappa\, i}, \qquad \kappa = 0.0005/\text{day}$$

The per-layer folding factor:

$$F_i = Q_i \cdot (26!)^{-1/13} \cdot S_{26}^{(3)} \cdot \Phi_{\text{gauss}}$$

The aggregate folding is the product over all layers:

$$\prod_{i=1}^{26} (1 + F_i)$$

4. Wolfram-Parallel Hypergraph Structure

The 26D folding maps to a hypergraph rewriting system:

- **Nodes**: 26 quantum layers (indexed $i = 1, \ldots, 26$)
- **Nearest-neighbor edges**: $i \to i + 1$ (ring topology, 26 edges)
- **Skip connections**: $i \to i + 13$ (modular skip, 26 edges)
- **Total edges**: 52
- **Graph density**: $52/\binom{26}{2} = 52/325 \approx 0.16$

Each node update applies the folding operator $\mathcal{F}_{26}$, analogous to Wolfram's rule application on spatial hypergraph nodes. Computational irreducibility emerges from the nonlinear interaction between $S_{26}^{(3)}$ and the phonon spectral function.

5. Implementation

Calculator: `UQFF26DGeometricFoldingOperatorCalculator` in CondensedPhysics.py

- Inputs: field value x, radial coordinate r, mass M, phonon parameters
- Outputs: $\mathcal{F}_{26}(x)$, $\mathcal{F}_{26}(r)$, folded metric components, $r_{s,\text{folded}}$, hypergraph statistics, layer factors

References

- Wolfram, S. (2020). A class of models with the potential to represent fundamental physics. *Complex Systems* 29(2).
- Murphy, D.T. (2025). UQFF: Star Cradle Mechanics framework. Star-Magic repository.
- PAPER_624: UQFF 26D Simultaneous Geometric Infinity Sculpting.